

Graphs clustering using Self organising maps

Some data is naturally a graph. A molecule is a set of atoms joined by bonds, a program is a set of statements joined by control flow. When you have a collection of such objects and want to know whether they fall into families, you need a way of saying how similar two graphs are. Unlike two vectors, two graphs have no natural alignment, since there is no reason to say that one graph’s third node corresponds to another’s, so every method works by summarising each graph and comparing the summaries. This thesis studies four such summaries, the graph edit distance, a spectral distance, a Weisfeiler Leman distance and a shortest path distance. The clustering method is a self organising map, which arranges the graphs on a grid so that similar ones end up near each other.

The usual way to choose between distances is to try each one, cluster, and keep whichever scores best. That comparison cannot work. A clustering score is produced jointly by three things, what the distance can represent, how far apart the classes happen to be, and how good the algorithm is, and changing any one of them moves the score. A low score might mean the distance is blind to the property, or that the dataset barely separates it, or that the algorithm failed. These are different conclusions, and the experiment cannot tell them apart.

The problem runs deeper, because the confusion comes from how the dataset is built. Graph properties travel together, since graphs with more edges also have more triangles, higher degrees and smaller diameters. Choosing graphs by one property means choosing them, in part, by all the others. So when a distance separates the classes, you cannot say what it responded to.

Instead of asking the three questions at once, it measures them separately.

First, what each distance can represent, using no dataset and no clustering at all. We list every graph of a given size and count how often each distance gives two different graphs exactly the same value. We take graphs drawn at random and work out which elementary properties the distance is actually a function of. And we take a single graph, change one property by one unit while keeping everything else fixed, and see whether the distance notices.

Second, how far apart the classes lie in a given dataset, measured before any algorithm runs.

Third, what the clustering achieves, comparing self organising maps against k medoids and hierarchical clustering on problems whose difficulty we generate deliberately and then measure, so that the algorithms are judged on equal terms.

Designed datasets are then used in only one way, to test predictions written down in advance from the first stage, which used no designed data at all. A dataset built to demonstrate something proves little, while a dataset used to check a prediction formed elsewhere is a real test.

The four distances are far more different from one another than their definitions suggest. The shortest path distance cannot tell apart 99.6% of eight node graphs, mapping over eleven thousand graphs onto fewer than two hundred values, and yet it was the most successful of the four at clustering, because the little it keeps includes exactly the property being looked for. The graph edit distance never confuses two graphs and is by far the most expensive, but under unit costs it returns the same number for any two edits of equal cost, whatever they do to the graph, and it clustered poorly. What predicts success is not how much a distance keeps, but whether what it keeps is what the task needs.

A property can be present in a representation and still be out of reach. The number of connected components is exactly the number of zero eigenvalues of a graph’s Laplacian, so the spectral

distance encodes it with mathematical certainty. On a dataset where everything else was held equal, it recovered almost none of it, while a cheaper distance recovered it easily.

How far apart the classes lie explains most of what happens next. Across two very different kinds of dataset, separation measured before clustering predicted recovery afterwards almost perfectly, while the choice among four clustering algorithms explained under one percent of the outcome. On this evidence the distance is not a setting to be tuned but a decision about what counts as similar, and given the same graphs with no labels, the three cheap distances organise them in three different ways.

The self organising map did not cluster better than simpler methods. What it adds is a layout that can be inspected, and how compactly a class sits on the grid turns out to carry real information about whether the clustering worked.

Finally, applying all of this to the chromatic number, a property none of these distances was designed for, every distance failed once graph size was controlled, while a single elementary statistic separated the same classes easily. That failure had been predicted in advance, from an experiment that never touched a colouring problem.

The practical message is that the choice of graph distance can be made before any clustering is run, by measuring what the distance responds to, and that this measurement predicts what will happen afterwards. The methodological message is that a graph clustering result is only interpretable when the difficulty of the problem has been measured rather than assumed. Along the way the thesis reports three results that went against its own expectations, including one of its own datasets that turned out to test something other than what it was built for.